

Assignment 1 – Relational databases

Description

We will work with the Yelp dataset: <https://business.yelp.com/external-assets/files/Yelp-JSON.zip>. You must unzip the file; inside the “Yelp JSON” folder, you will find two files. The first file is a PDF file describing the dataset. Note that this description is not complete and you must explore the data files. The second file is named “yelp_dataset.tar”. You must rename this as “yelp_dataset.tar.gz”, and this is the file your programs will receive as input. To work with the file, do not decompress it; instead, read it programmatically as shown in the software templates provided. The templates have additional hints you must follow.

The database system we are going to use is MySQL Community Server version 8.0.44. The Java and Python versions are 21 and 3.12, respectively.

Your tasks

1.- Provide a program that loads the business. You must create and populate with data the following relations into MySQL:

- Business(id, name, address, zc_id, latitude, longitude, is_open, bike_parking, good_for_dancing, restaurants_good_for_groups, restaurants_price_range, restaurants_take_out, restaurants_delivery, restaurants_counter_service, restaurants_table_service, restaurants_reservations, drive_thru, wheelchair_accessible, by_appointment_only, dogs_allowed, happy_hour, byob, corkage, outdoor_seating, accepts_insurance, caters, coat_check, good_for_kids, open_24_hours, has_tv, business_accepts_credit_cards, business_accepts_bitcoin, PK(id), FK zc_id TO Zipcode(id))
- Zipcode(id, code, city_id, PK(id), FK city_id TO City(id))
- City(id, name, state_id, PK(id), FK state_id TO State(id))
- State(id, name, PK(id))
- Category(id, name, PK(id))
- BusinessCategory(bid, cid, PK(bid, cid), FK bid TO Business(id), FK cid TO Category(id))
- BusinessHour(bid, day, open_time, closing_time, PK(bid, day), FK bid TO Business(id))

You can check the expected types in the MySQL data dump provided and other issues you may encounter. There are memory and time requirements. Check the grading software and software templates for more info.

2.- Provide a program that loads the users. You must create and populate with data the following relations into MySQL:

- User(id, name, yelping_since, useful, funny, cool, fans, compliment_hot, compliment_more, compliment_profile, compliment_cute, compliment_list, compliment_note, compliment_plain, compliment_cool, compliment_funny, compliment_writer, compliment_photos, PK(id))
- UserElite(uid, year, PK(uid, year), FK uid TO User(id))

You can check the expected types in the MySQL data dump provided and other issues you may encounter. There are memory and time requirements. Check the grading software and software templates for more info.

Next, we will work with SQL. To obtain the expected results, you must use the MySQL data dump available in myCourses, that is, you must not rely on your own data but use the common data dump. The MySQL server must use the `ONLY_FULL_GROUP_BY` option (<https://dev.mysql.com/doc/refman/8.0/en/group-by-handling.html>). You can test this by trying a GROUP BY query without a GROUP BY clause. The server must output an error.

The schema uses the previous relations as well as this one:

- Review(id, uid, bid, stars, review_date, useful, funny, cool, PK(id), FK uid TO User(id), FK bid TO Business(id))
- Friend(uid_one, uid_two, PRIMARY KEY(uid_one, uid_two), FK uid_one TO User(id), FK uid_two TO User(id))

Notes:

- We will consider the friend relation to be directed. If you are looking for the friends of user u7, you will get the results such that uid_one is u7. You should not use uid_two to search for the friends of a given user.
- Sunday is Day 0.

3.- Provide a single SQL query to retrieve the following data:

3.1.- Open businesses that have received more than two hundred and fifty reviews, that belong to more than five categories, and that are closed on Saturdays and Sundays.

3.2.- Top-15 most prolific users who are friends of more than two-hundred and fifty other users and have not reviewed any Mexican businesses. Reviews must be grouped by category.

3.3.- Reviews written by users who started yelping in June 2015 such that there are posterior reviews by the same user, with the exact same stars as well as funny, cool and useful votes. The posterior reviews were written on January 1st, 2020. Retrieve (original id, posterior id) pairs.

3.4.- Top-5 users such that they have reviewed at least five Bookstores with four or more stars. The users must have started yelping after 2021.

3.5.- Users whose name contains either "ch" or "th" (case insensitive), who have reviewed more than two businesses that belong to categories Korean or Seafood, who have no friends, and who have not reviewed any Mexican businesses after 2020.

3.6.- Users who started yelping between January 2021 and May 2021, who have reviewed at least three businesses in Arizona, who are friends of at least two other users, and who have never been elite users.

3.7.- Users with at least seventy-five friends, and who have reviewed at least one business such that the review got more than three funny and cool votes. The users started yelping strictly after all their friends started yelping.

3.8.- Users with more than 50 fans and more than five friends, and all their friends have written more than three five-star reviews. The users have never reviewed a business located in Largo.

3.9.- Businesses reviewed by all the reviewers of Bayou Daiquiris and the average number of stars received. The average number of stars must be cast as CHAR.

3.10.- Korean businesses that, during weekdays, are open at the exact same times or longer as Bayou Daiquiris. For the Korean businesses, provide the total number of useful votes. The total number of useful votes must be cast as CHAR.

There are memory and time requirements. Check the grading software and query templates for more info.

Submission instructions

- Use the software template(s) and grading software provided in myCourses.
- Submit a single ZIP file to myCourses that must be named as your RIT user, e.g., `errvcs.zip`. Do not include '@rit.edu.' The file must contain folders named as the folders provided in the software template(s) containing your queries and/or Gradle/Poetry project(s).
- Everything will be graded on a Linux machine, so you must always use the exact names provided in this document, software and query template(s), and grading software.

Grading rubric

- All test cases in the grading software must pass with some restrictions like main memory and total running time.